

Executing SQL Queries



Hands-on Lab : Executing SQL Queries

Estimated time needed: 30 minutes

In this lab you will be using phpMyAdmin, which is a free tool embedded in this lab environment to work on MySQL.

Objectives

After completing this lab you will be able to:

- Create a Database
- Create and load tables using csv files
- Execute SQL queries

Software used in this lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.

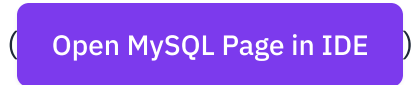


To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

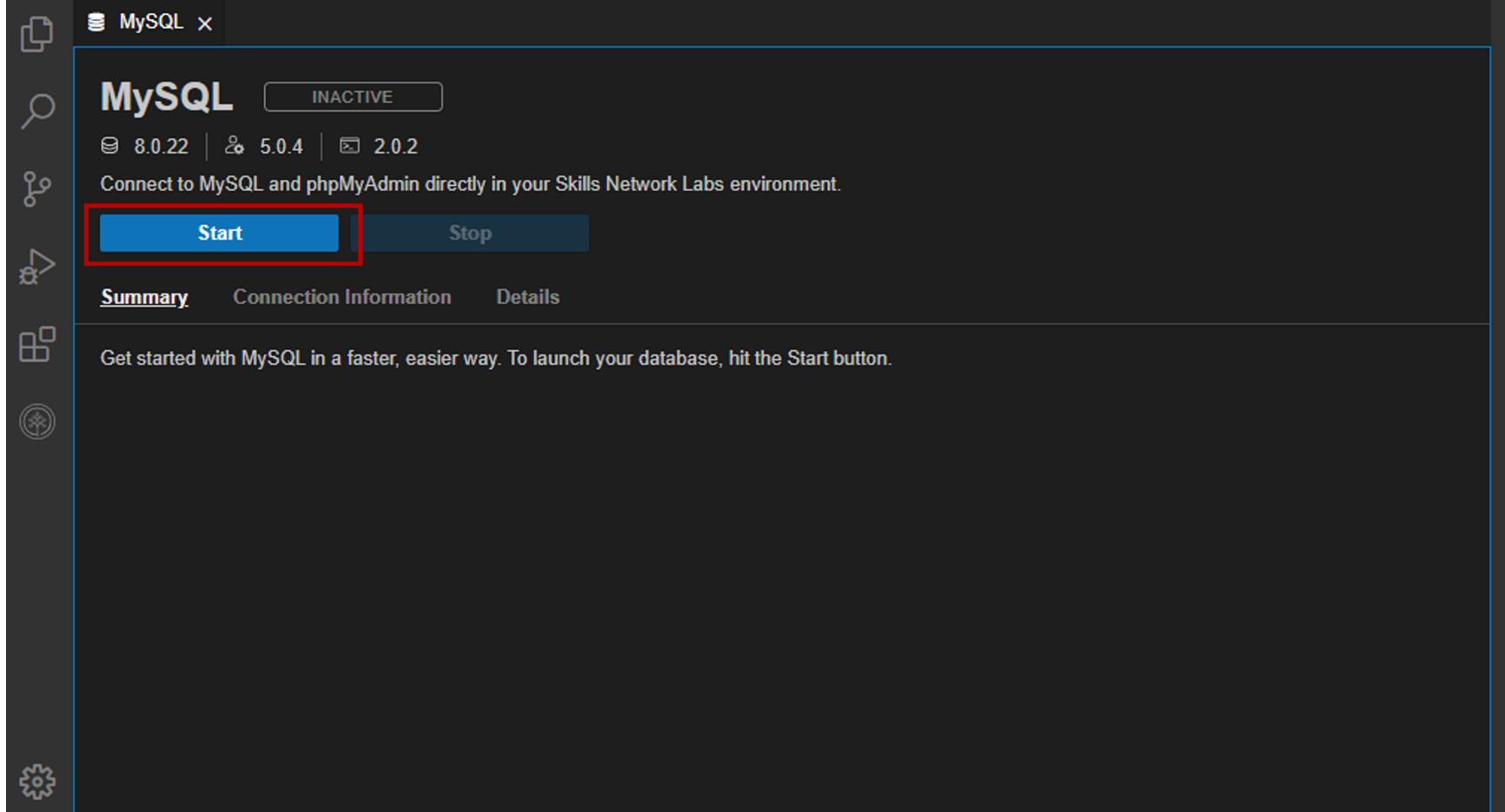
Prework - Create and populate database

TASK A: Create a Database

1. Start the MySQL service session using the [Open MySQL Page in IDE](#) button.



To start the MySQL, click [Start](#).



2. Once MySQL has started, click on phpMyAdmin button to open phpMyAdmin in the same window.

The screenshot shows the Skills Network Labs interface for a MySQL database. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. On the left sidebar, there are icons for various categories: SKIL..., DATABASES, BIG DATA, CLOUD, OTHER, and a Launch A... option. The main panel displays the MySQL status as 'ACTIVE' in a green box. Below this, it shows versions for MySQL (v8.0.22), phpMyAdmin (v5.0.4), and a terminal icon (v14.14). A message states: 'Connect to MySQL and phpMyAdmin directly in your Skills Network Labs environment.' A blue 'Stop' button is visible. Below the message, there are three tabs: 'Summary' (selected), 'Connection Information', and 'Details'. The 'Summary' tab contains the text: 'Your database and phpMyAdmin server are now ready to use and available with the following login credentials. For more details on how to navigate MySQL, please check out the Details section.' The credentials listed are: Username: lavanyas and Password: NzgyNS1sYXZhbnlh. Below the credentials, it says 'You can manage MySQL via:' followed by a blue button labeled 'phpMyAdmin' and a red-outlined square button containing a white icon of a document with an arrow pointing out, which is the link to the phpMyAdmin GUI.

File Edit Selection View Go Run Terminal Help

SKIL...

> DATABASES

> BIG DATA

> CLOUD

> OTHER

Launch A...

MySQL **ACTIVE**

v8.0.22 | v5.0.4 | v14.14

Connect to MySQL and phpMyAdmin directly in your Skills Network Labs environment.

Stop

Summary Connection Information Details

Your database and phpMyAdmin server are now ready to use and available with the following login credentials. For more details on how to navigate MySQL, please check out the Details section.

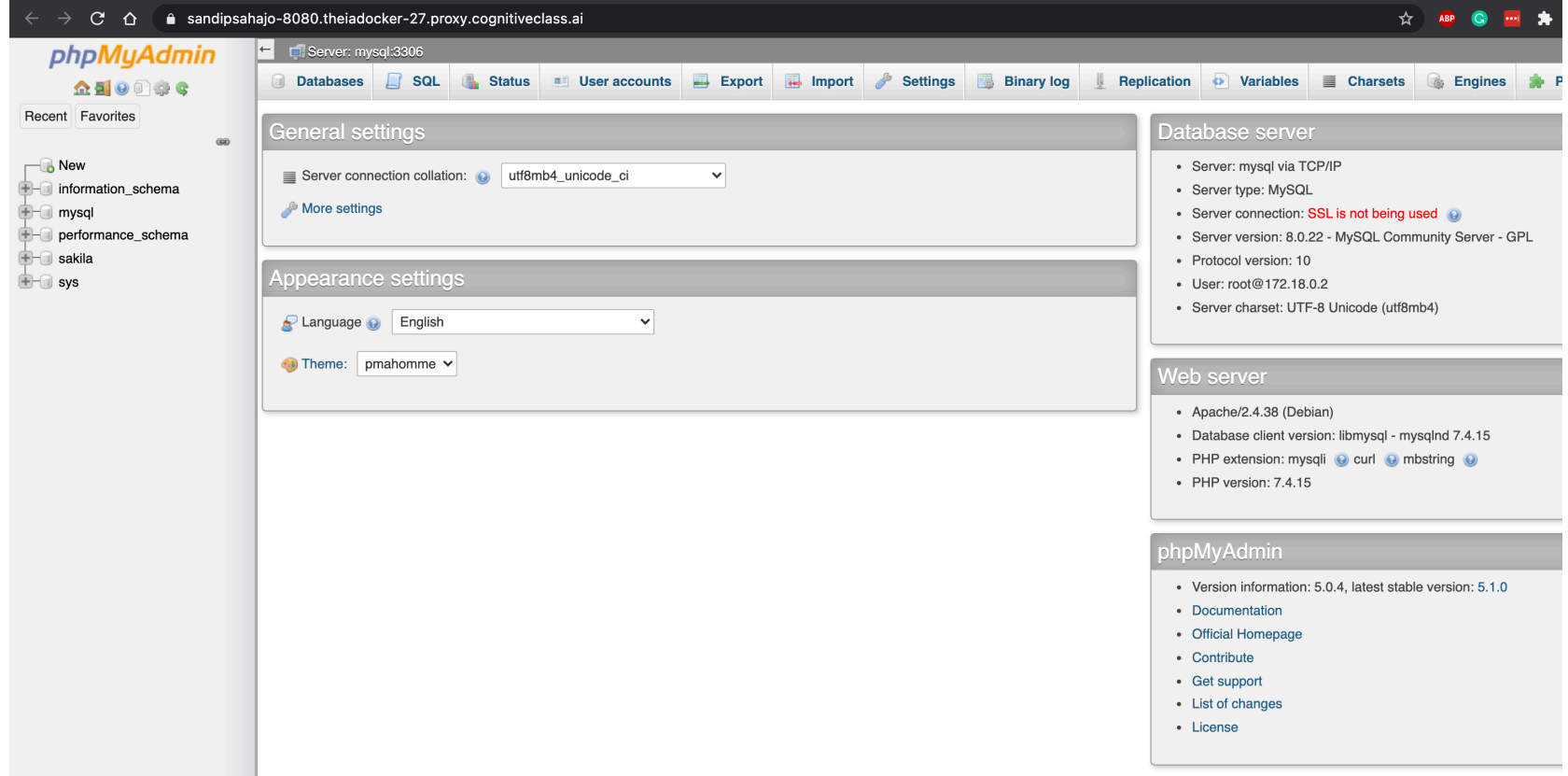
Username: lavanyas

Password: NzgyNS1sYXZhbnlh

You can manage MySQL via:

phpMyAdmin

3. You will see the phpMyAdmin GUI tool.



4. In the tree-view, click **New** to create a new empty database. Then enter **Mysql_Learners** as the name of the database and select utf8_general_ci and click **Create**.

UTF-8 is the most commonly used character encoding for content or data.

[Databases](#)[SQL](#)[Status](#)[User accounts](#)[Export](#)[Import](#)

Databases

[Create database](#)[Create](#)

	Database	Collation	Master replication	Action
☐	information_schema	utf8_general_ci	✓ Replicated	Check privileges
☐	mysql	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
☐	performance_schema	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
☐	sys	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
Total: 4				

TASK B: Create and load tables using csv files.

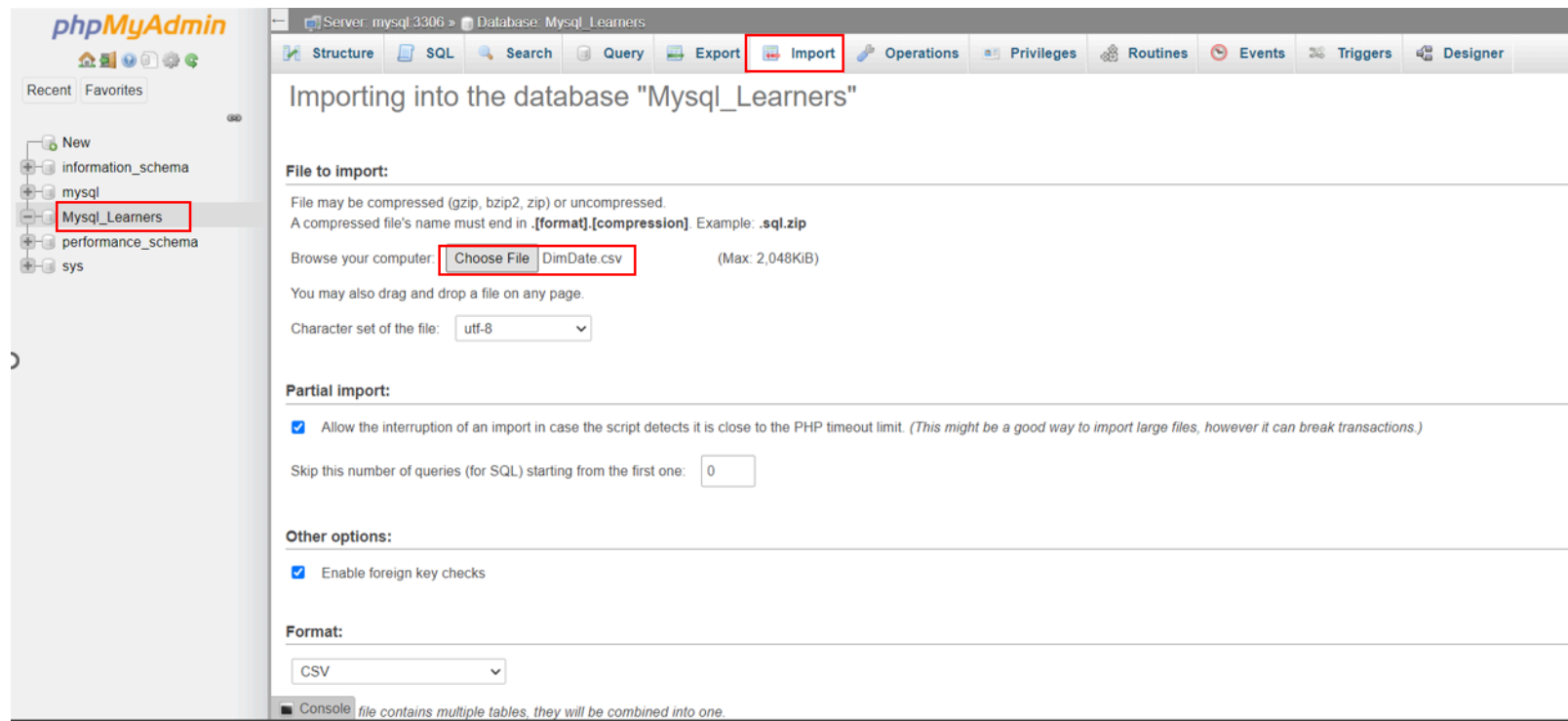
1. Download the 4 csv files below to your local computer:

- [dimdate.csv](#)
- [dimtruck.csv](#)
- [dimstation.csv](#)

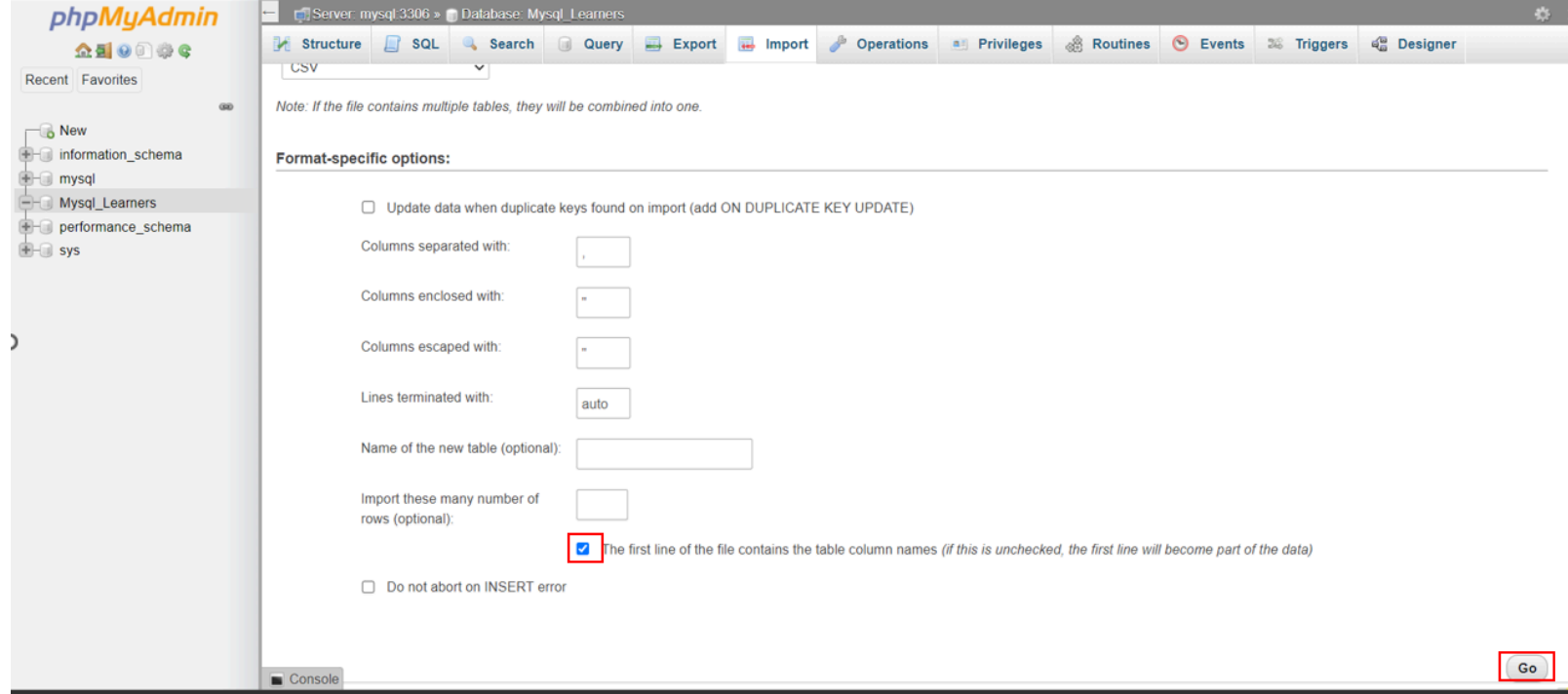
- [facttrips.csv](#)

2. To load each **csv** file do the following steps.

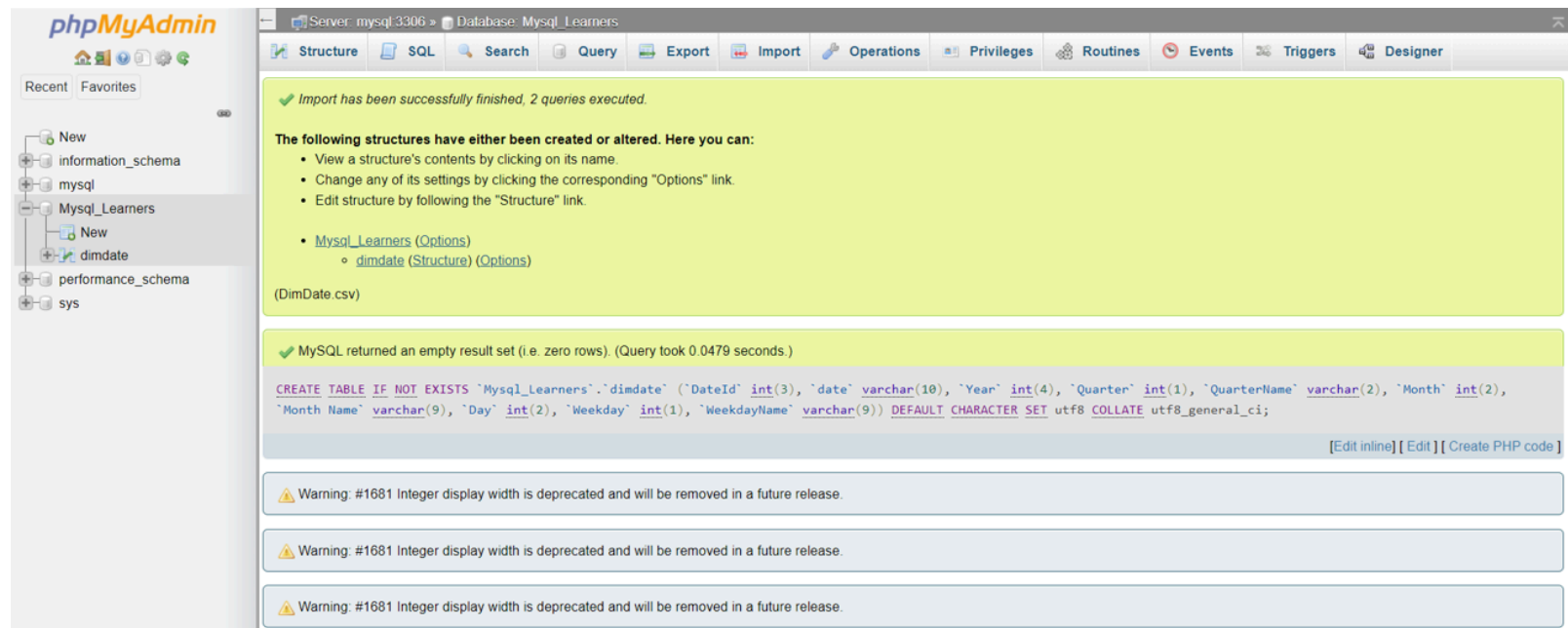
- Select your database **Mysql_Learners** and click on **Import** tab and select the **csv** file.



- Then scroll down and check the box as shown below and click on **Go** to load the csv file.

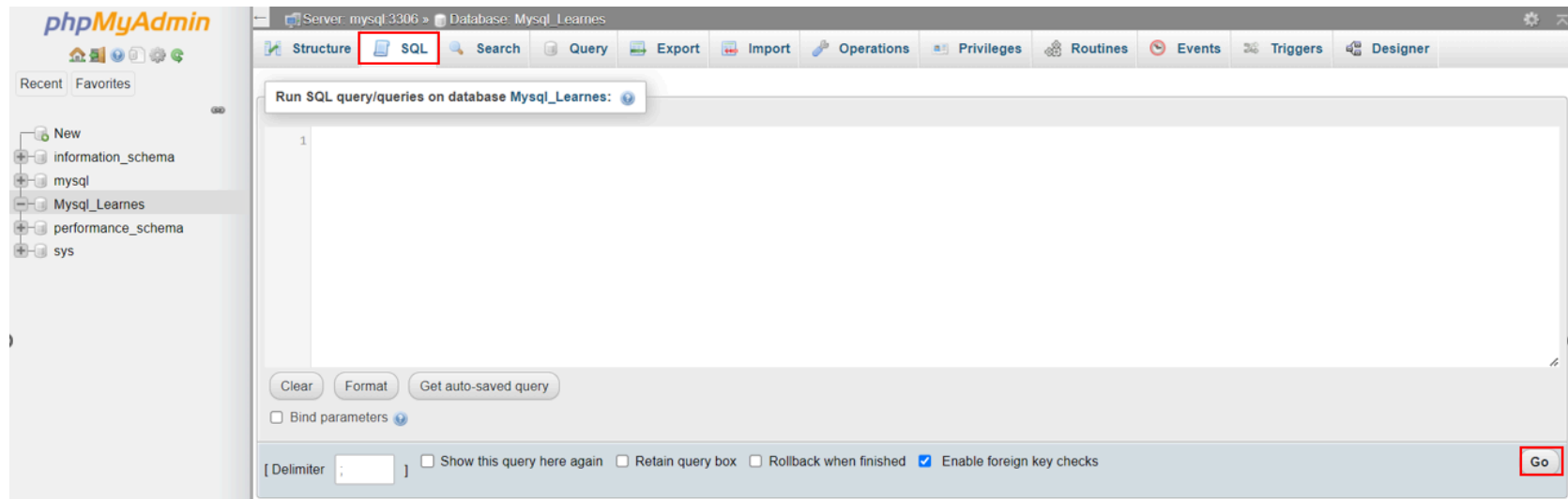


- Once the tables are loaded , you will get a message that the records are inserted successfully.



Further You can import all the other csv files in the same way.

3. To run the SQL queries you need to copy the given codes and paste it to the text area of the SQL page and click on **Go**.



Execute SQL Queries

Exercise 1: List all stations in an alphabetical order. Output should contain `StationId`, `StationName`.

► Solution Syntax

► Output

Exercise 2: List all trips that collected waste > 40. Output should contain `TripId`, `Waste`.

► Solution Syntax

► Output

Exercise 3: List average waste collected for each date. Output should contain `DateId`, `average waste`.

► Solution Syntax

► Output

Exercise 4: List truck Names with their count. Output should contain **TruckName, count**

▶ Solution Syntax

▶ Output

Exercise 5: List City with total waste collected. Output should contain **CityName, total_Waste**

▶ Solution Syntax

▶ Output

Exercise 6: List minimum waste collected per quarter in 2019. Output should contain **QuarterName, minimum waste.**

▶ Solution Syntax

► Output

Exercise 7: List maximum waste collected in Q1 in Sao Paulo. Output should contain QuarterName, City, maximum Waste.

► Solution Syntax

► Output

Exercise 8: List the days of the week results in the highest average waste collected by Volvo trucks. Output should contain WeekDayName, TruckName, avg_Waste.

► Solution Syntax

► Output

Exercise 9: List the dates when each city collected its maximum Waste. Output should contain city, date, maximum Waste.

► Solution Syntax

Congratulations! You have completed this lab successfully.

Authors

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Other Contributors

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Prework - Create and pop... →